

PROFILE

- Operations Research graduate student at IIT Kanpur with experience in **Supply Chain Analytics, Logistics Optimization, Network Design, Demand Forecasting, Machine Learning, and GenAI Applications** through academic and industry projects in manufacturing, logistics, and analytics.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. IME	Indian Institute of Technology, Kanpur	9.46	2024-2026
B.Tech. Mechanical Engineering	J.C.Bose University, YMCA, Faridabad	8.6	2018-2022
Senior Secondary	HBSE Board	94%	2018
Secondary	HBSE Board	95%	2016

RESEARCH EXPERIENCE (M.TECH. THESIS)

Aug'25 – May'26

- Pollution Routing Problem (PRP): Sustainable Logistics Optimization | Supervisor: Dr. Faiz Hamid**
 - Analyzed trade-off between transportation cost, delivery time, and emissions in **logistics routing decisions**.
 - Built foundational B&P framework on **VRPTW** using **Column Generation** for master problem decomposition and **Dynamic Programming** (labeling algorithms with **2-cycle elimination**) for the pricing subproblem.
 - Achieved **optimal solutions** on **Solomon benchmarks (100 nodes)**, **3.7× faster** than direct MILP formulation in Gurobi.
 - Extending to PRP by incorporating **speed as a decision variable**, **load-dependent fuel models**, and **CO₂ emission constraints** in the pricing subproblem.

WORK EXPERIENCE

- L&T Finance | AI/ML Intern | Risk Analytics (Internship)**

May'25 – Jul'25

 - Developed **state-level credit risk models** using macroeconomic indicators and feature engineering to predict default risk.
 - Analyzed large customer datasets** and validated models using statistical diagnostics to ensure robust risk predictions.
 - Built a **spatio-temporal location stability model** using neural network embeddings of GPS data to identify reliable locations.
 - Improved customer-location match accuracy by **13%** using clustering techniques (**K-Means**).
- Maruti Suzuki India Limited | Assistant Manager | Operations & Data Analysis (Full-Time)**

Jul'22 – Jul'24

 - Analyzed machine sensor data to identify performance trends and support **operational efficiency improvements**.
 - Analyzed **sensor data using Excel** to derive performance insights and recommended **design modifications** to the engineering team, improving fuel efficiency by **1.2%**.
 - Improved manufacturing throughput by reducing bottleneck machine downtime by **10%** through **root-cause analysis**.

PROJECTS

- Pricing Strategy Optimization using Predictive & Prescriptive Analytics | Course Project**

Github July'25

 - Developed a **data-driven pricing strategy** using demand forecasting and optimization models to support revenue decision-making.
 - Built a **demand prediction model ($R^2 \approx 0.90$)** capturing price elasticity, seasonality, and regional demand patterns.
 - Formulated a **pricing optimization model** in **Gurobi** incorporating revenue, transportation costs, and wastage trade-offs.
 - Identified optimal pricing and supply allocation strategies leading to **~35% higher net revenue** compared to historical benchmarks.
- Supply Network Optimization (MILP – Gurobi, Python) | OR Course Project**

Github Dec'24

 - Developed an optimized **supply chain model** to minimize **network cost** while meeting **regional demand**.
 - Formulated the problem as a **MILP** using **Python–Gurobi**, modeling factories, depots, and customer flows.
 - Incorporated **supply limits, demand constraints, and cost parameters** with binary and continuous decision variables.
 - Performed **sensitivity analysis** on transport, expansion costs and demand scaling to evaluate **network robustness**.
- Predictive Maintenance Analytics for Railway Infrastructure | ML | Kaggle Competition**

Github Nov'25

 - Developed a **predictive analytics model** to identify **5 railway wheel failure types** using **20M+ multi-sensor observations**.
 - Engineered domain-specific features and addressed class imbalance using **SMOTE** to improve failure detection accuracy.
 - Benchmarked **LSTM, XGBoost, Random Forest, and LightGBM** with **Optuna hyperparameter tuning** to identify the best-performing model.
 - Achieved **Rank 3** in this competition and presented results at the **INFORMS Conference** and **International Rail Conference**.
- Website Assistant using Advanced RAG | LLMOps | GenAI | Self Project**

Github May'25

 - Built an **AI-powered system** for natural language interaction with websites using **RAG & LLMOps**.
 - Integrated **OpenAI Embeddings, Pinecone Vector DB, Hybrid Search, and Cohere re-rankers** for optimized retrieval.
 - Performed **RAG evaluation** with **RAGAS framework**, achieving **84% faithfulness** and **0.78 relevance** for 100 samples.

RELEVANT COURSEWORK

- Operations Research | Advanced Decision Modeling | Statistical Modeling for Business Analytics | Applied Machine Learning | Game Theory | Probability & Statistics

TECHNICAL SKILLS & CERTIFICATIONS

- Programming:** Python, SQL, L^AT_EX
- Supply Chain & Analytics:** Supply Chain Network Design, Logistics Optimization, Demand Forecasting, Prescriptive Analytics
- Data Analysis & Visualization:** Excel / Google Sheets, Pandas, NumPy, Matplotlib
- Tools & Solver:** AWS, GCP, Gurobi, CPLEX, Pyomo, Mathematical Programming, Optimization Modeling
- Certifications:** Large Scale Optimization Summer School (LSO 2025), IIT Bombay

ACHIEVEMENTS

- Academic Excellence Award** Department of Management Sciences | IIT Kanpur

Jan'25
- AIR 7** in GATE XE 2024 organized by IISc Bengaluru

Apr'24
- Received NTPC offer letter** for the post of Executive Trainee on the basis of GATE 2024

July'25